



Cambridge International AS & A Level

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FURTHER MATHEMATICS

9231/22

Paper 2 Further Pure Mathematics 2

May/June 2026
2 hours

You must answer on the question paper.

You will need: List of formulae (MF19)

INSTRUCTIONS

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen. Do **not** use correction fluid or tape.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined pages at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

INFORMATION

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [].

 This document has **16** pages.

- [illegible]



2 Let $I_n = \int_0^1 (1+t)^n \sin \pi t \, dt$, where n is a non-negative integer.

(a) Show that, for $n \geq 2$, $\pi^2 I_n = \pi(2^n + 1) - n(n-1)I_{n-2}$. [4]

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(b) Find the exact value of I_2 . [3]

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[4]

[illegible]


$$\frac{dy}{dx} + y \coth x = \sinh^2 x,$$
[illegible]

4

$$(x-y)^5 + \ln(xy) = 1 + \ln 2.$$

(a) Show that, at the point $(-1, -2)$ on C , $\frac{dy}{dx} = \frac{8}{11}$. [3]

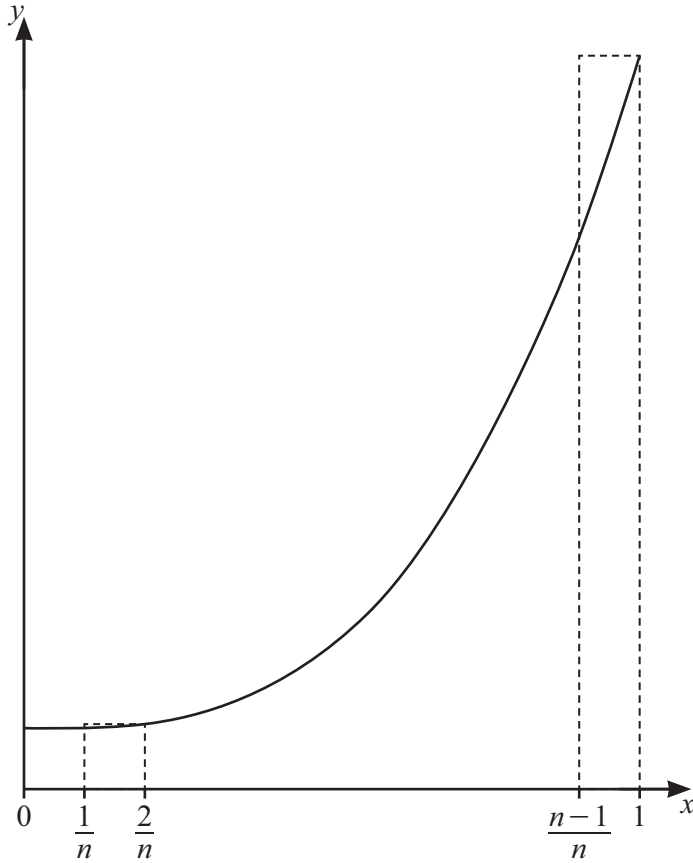
This image shows a full page of a document template. It consists of approximately 28 horizontal dotted lines spaced evenly down the page, providing a guide for handwriting or typing. The background is plain white, and there are no margins, headers, or footers visible.



[5]

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

5



The diagram shows the curve with equation $y = \frac{11}{10}x^3 + \frac{1}{10}$ for $0 \leq x \leq 1$, together with a set of n rectangles of width $\frac{1}{n}$.

- (a) By considering the sum of the areas of these rectangles, show that $\int_0^1 \left(\frac{11}{10}x^3 + \frac{1}{10} \right) dx < U_n$, where

$$U_n = \frac{11}{40} \left(1 + \frac{1}{n} \right)^2 + \frac{1}{10}. \quad [4]$$

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- (b) Use a similar method to find, in terms of n , a lower bound L_n for $\int_0^1 \left(\frac{11}{10}x^3 + \frac{1}{10} \right) dx$. [4]

- (c) Show that $\lim_{n \rightarrow \infty} (U_n - L_n) = 0$. [2]



6

$$5\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + y = x^2 + 7x + 1,$$

given that, when $x = 0, y = 0$ and $\frac{dy}{dx} = 0$.

[10]

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[illegible][illegible]

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8

$$ax + y - z = 2,$$

$$by + 2z = 3$$

[3]

The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{pmatrix} -1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 2 \end{pmatrix}.$$

(b) Use the characteristic equation of $\mathbf{A}$ to show that $\mathbf{A}^4 = p\mathbf{A}^2 + q\mathbf{I}$, where p and q are integers to be determined. [4]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.



[7]

[illegible]

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